

Social Cognitive Theory and Motivation

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Abstract

Social cognitive theory is a theory of psychological functioning that emphasizes learning from the social environment. This chapter focuses on Bandura's social cognitive theory, which postulates reciprocal interactions among personal, behavioral, and social/environmental factors. Persons use various vicarious, symbolic, and self-regulatory processes as they strive to develop a sense of agency in their lives. Key motivational processes are goals and self-evaluations of progress, outcome expectations, values, social comparisons, and self-efficacy. People set goals and evaluate their goal progress. The perception of progress sustains self-efficacy and motivation. Individuals act in accordance with their values and strive for outcomes they desire. Social comparisons with others provide further information on their learning and goal attainment. Self-efficacy is an especially critical influence on motivation and affects task choices, effort, persistence, and achievement. Suggestions are given for future research directions.

Key Words: social cognitive theory, vicarious processes, symbolic processes, self-regulatory processes, goals, self-evaluations of progress, outcome expectations, values, social comparisons, self-efficacy

Introduction

Motivation refers to the process whereby goal-directed activities are energized, directed, and sustained (Schunk, Pintrich, & Meece, 2008). Contemporary cognitive theories of motivation postulate that individuals' thoughts, beliefs, and emotions are central processes that underlie motivation. These cognitive perspectives stand in contrast both to early views that linked motivation with individual differences in instincts and traits and to behavioral theories that viewed motivation as an increased or continued level of responding to stimuli caused by reinforcements or rewards.

In this chapter we provide an account of motivation from the perspective of social cognitive theory. *Social cognitive theory* emphasizes the idea that much human learning and behavior occur in social environments.

By interacting with others, people learn knowledge, skills, strategies, beliefs, rules, and attitudes. Through their observations and interactions with others, individuals also learn about the appropriateness, usefulness, and consequences of behaviors. People act in accordance with their beliefs about their capabilities and the expected outcomes of actions.

Although there are different social cognitive perspectives on motivation, this chapter focuses on Bandura's (1977b, 1986, 1997, 2001) social cognitive theory of psychological functioning. Bandura's theory underscores the key roles of vicarious, symbolic, and self-regulatory processes in human learning and behavior. This social cognitive framework often is employed by researchers to explore the operation and outcomes of cognitive and affective processes hypothesized to underlie motivation.

The next section describes the conceptual framework of social cognitive theory to include the key roles played by vicarious, symbolic, and self-regulatory processes. Some key social cognitive motivational processes are discussed, including goals and self-evaluations of progress, outcome expectations, values, social comparisons, and self-efficacy. A separate section is devoted to discussing *self-efficacy*, or one's perceived capabilities for learning or performing actions at designated levels (Bandura, 1977a, 1997), given its centrality to learning and motivation. We conclude the chapter with suggestions for future research.

Conceptual Framework

This section discusses the conceptual framework of social cognitive theory. Of particular importance are the following: reciprocal interactions among personal, behavioral, and social/environmental factors; the differences between enactive and vicarious learning; the distinction between learning and performance; and the roles of vicarious, symbolic, and self-regulatory processes in psychological functioning.

Reciprocal Interactions

A central tenet of Bandura's (1977b, 1986, 1997, 2001) social cognitive theory is that human behavior operates within a framework of *triadic reciprocity* involving reciprocal interactions among three sets of influences: personal (e.g., cognitions, beliefs, skills, affect); behavioral; and social/environmental factors. This reciprocal network is illustrated in Figure 2.1.

These reciprocally interacting influences can be demonstrated using self-efficacy as the personal

factor. With respect to the interaction of self-efficacy and behavior, much research shows that self-efficacy influences achievement behaviors such as task choice, effort, persistence, and use of effective learning strategies (person → behavior; Schunk & Pajares, 2009). These behaviors also affect self-efficacy. As students work on tasks and observe their learning progress, their self-efficacy for continued learning is enhanced (behavior → person).

The link between personal and environmental factors can be illustrated with students with learning disabilities, many of whom hold low self-efficacy for performing well (Licht & Kistner, 1986). Instructors in such environments may base their reactions to these students on perceived attributes about the students (e.g., low skills) rather than on students' actual capabilities (person → social/environment). In turn, environmental feedback can affect students' self-efficacy, such as when a teacher tells a student, "I know you can do this" (social/environment → person).

The link between behaviors and environmental factors is seen in many instructional sequences. Environmental factors can direct behaviors, such as when a teacher points to a display and says, "Look here," which students do without much conscious effort (social/environment → behavior). Students' behaviors can alter their instructional environments. When teachers ask questions and students give incorrect answers, teachers are apt to reteach the material rather than continue with the lesson (behavior → social/environment).

Social cognitive theory reflects a view of human *agency* in which individuals are proactively engaged in their own success and development (Schunk &

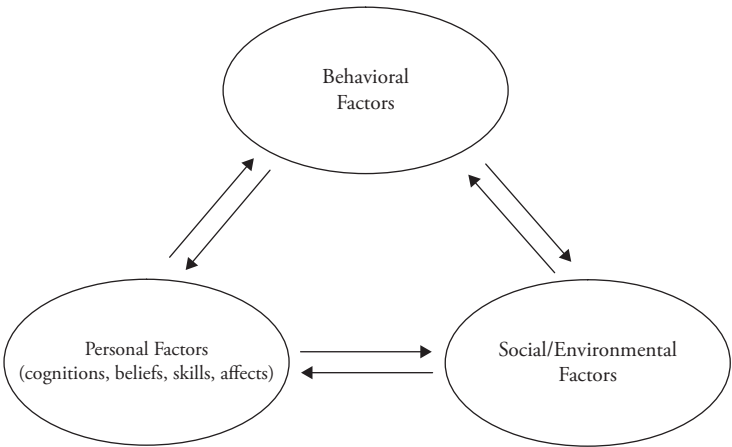


Fig. 2.1. Reciprocal interactions in social cognitive theory.

Pajares, 2005). These beliefs allow individuals to exert a large degree of control over their thoughts, feelings, and actions. In reciprocal fashion, people affect and are influenced by their actions and environments. But the scope of this reciprocal influence is broader than individuals because they live in social environments. *Collective agency* refers to people's shared perceived capabilities of accomplishing tasks as a group. As is true with individuals, groups also affect and are influenced by their actions and environments.

Enactive and Vicarious Learning

In social cognitive theory, learning occurs *enactively* through actual doing and *vicariously* through observing modeled performances (e.g., live, filmed, symbolic; Bandura, 1977b). Enactive learning involves learning from the consequences of one's actions, which can inform and motivate. Actions convey information about the accuracy or appropriateness of one's behaviors. People rewarded for their actions typically understand that they are performing well, whereas punishments signal behavioral inappropriateness. Individuals tend to be motivated to learn and perform behaviors that they believe will have desirable consequences and to avoid learning behaviors that they believe will be punished.

Much human learning occurs vicariously and therefore does not require actual performance by learners. Vicarious learning offers an efficient alternative to learning via direct experience. Humans would be hopelessly inefficient if their involvement were required for all learning. Vicarious learning also saves people from undesirable consequences. Observing or reading about safety techniques saves individuals from acting in potentially dangerous ways. As with enactive learning, observers are motivated to learn actions that lead to successes. People attend to successful models who demonstrate actions that they believe will benefit them (Schunk, 1987).

Learning of complex skills typically occurs both enactively and vicariously. By observing teacher models, students may learn some aspects of a complex skill. As students practice the skills, teachers provide feedback and corrective instruction as needed. Through observation, practice, and feedback, students learn skills and enjoy greater success.

Learning and Performance

Unlike older behavioral theories, social cognitive theory distinguishes new learning from performance of previously learned actions (Bandura, 1977b). The

distinction is not apparent with enactive learning because persons demonstrate what they have learned. But vicarious learning may not be demonstrated until sometime after the modeled behavior occurs. Whether learning results in changed performances depends on factors such as learners' motivation, interests, incentives to perform, perceived needs, physical conditions, social pressures, and competing activities.

Students learn many new skills, strategies, and behaviors, only some of which they may demonstrate at the time of learning. Because teachers are responsible for ensuring that students learn, they assess student learning in various ways (e.g., tests, quizzes, assignments, homework). The assumption is that students will demonstrate what they have learned; however, this may not always happen. Able students who are motivated to be socially accepted by their peers may not demonstrate the full range of their learning so that they appear more in line with their classmates' competencies. Authentic assessments that take various forms can help teachers accurately gauge students' learning.

Vicarious, Symbolic, and Self-Regulatory Processes

Bandura's (1986) social cognitive theory stresses the idea that people possess capabilities that distinguish them as humans and motivate them to strive for a sense of agency. Among the most prominent of these are vicarious, symbolic, and self-regulatory processes.

Vicarious Processes

The capability for learning vicariously allows individuals to acquire beliefs, cognitions, affects, skills, strategies, and behaviors, from observations of others in their social environments. As noted earlier, this capability saves people time over what would be required if all learning had to be demonstrated at the time of learning. This capability also allows people to regularly shape their lives, because they select environmental features (e.g., individuals, materials) to which they want to attend. Thus, students who want to become teachers enroll in education programs and put themselves in situations where they can learn vicariously, such as by attending classes, observing and working with classroom teachers, and reading books and other materials. The models in individuals' environments serve as important sources of information and motivation. This section discusses the relevance of vicarious processes for learning and motivation.

TYPES OF VICARIOUS PROCESSES

Bandura (1986) distinguished three types of vicarious processes: response facilitation, inhibition and disinhibition, and observational learning (Table 2.1). *Response facilitation* refers to modeled actions that serve as social motivators for observers to act in the same fashion. Response facilitation effects are common in everyday life. An individual walking down a street who encounters a group of people looking in a store window may be motivated to stop and look in the window.

Response facilitation effects do not represent learning because people already know how to perform the actions. The behaviors of others motivate observers' actions. There is evidence that response facilitation effects can occur without conscious awareness (Chartrand & Bargh, 1999).

Inhibition and disinhibition effects result from models strengthening or weakening observers' tendencies to act in given ways. Inhibition can happen when models are punished for their actions, whereas disinhibition can result when models perform threatening or prohibited actions without negative consequences. Classroom misbehavior may be disinhibited when students observe other students misbehaving without being reprimanded by the teacher; a sudden reprimand may inhibit further misbehavior.

Like response facilitation, inhibition and disinhibition represent motivational effects on behavior, not new learning. A difference between these two categories is that, whereas response facilitation involves behaviors that are socially acceptable, inhibition and disinhibition typically involve actions that have moral or legal implications (e.g., breaking rules) or involve strong emotions (e.g., fears).

Observational learning through modeling occurs when observers perform behaviors that they had not learned prior to exposure to the models (Bandura, 1969). Observational learning has four component processes: attention, retention, production, and motivation.

Observational learning requires that observers attend to relevant features so that they can be perceived. Certain features of models and situations

command better attention. Observers are more motivated to attend to models who have status and credibility, such as teachers. Task features can affect attention, such as when teachers use bright colors, oversized features, and interactive materials. Attention also is affected by observers' beliefs about the functional value of the modeled behaviors. Modeled activities that observers believe are important and likely to lead to desirable outcomes motivate them to pay attention. Students' attention should be raised when teachers provide verbal markers, such as when they announce that the material they are about to cover will be on a test.

Retention involves cognitively organizing, rehearsing, coding, and transforming information for storage in memory. Relative to the other processes of observational learning, social cognitive theory devotes less attention to this process. Theorists and researchers in the information processing tradition have addressed this aspect in depth (Matlin, 2009).

The third process—production—involves translating cognitive conceptions of modeled actions into behaviors. Especially with complex behaviors, it often is the case that observers will learn only some features. Learners refine their skills through practice and feedback that may include additional modeling.

Motivation is a key process in observational learning because onlookers are more apt to attend to, retain, and produce those modeled actions that they believe are important. People are selective; they do not learn or perform everything that they observe. Rather, they attempt to learn those actions that they believe will lead to desirable outcomes and help them attain their goals, and they avoid those actions that they believe will result in dissatisfying outcomes. As they observe the actions of others, people form expectations about different outcomes, which are based on their observations of models and their own experiences. Their learning and performances are based in part on these expectations.

MODEL CHARACTERISTICS

Researchers have investigated the characteristics of effective models. For example, perceived similarity between models and observers can affect modeling (Schunk, 1987). Similarity in important ways serves as a source of information for determining behavioral appropriateness, forming outcome expectations, and assessing one's self-efficacy. Age similarity between model and observer is important for gauging behavioral appropriateness but

Table 2.1. Types of Vicarious Processes

• Response facilitation
• Inhibition/disinhibition
• Observational learning

less so for actual learning, which is enhanced more by models whom observers believe are competent. When competence and age similarity do not match (e.g., younger model is more competent than a same-age model), children are swayed more by the competent model. Peers can be effective models when children hold self-doubts about their learning or performance capabilities. Viewing a similar peer successfully perform a task may raise observers' self-efficacy and motivate them to learn because they are apt to believe that if the model could learn, they can as well (Schunk, 1987).

Model gender can influence modeling by conveying information about task appropriateness. In general, observing a same-gender peer model perform a behavior without negative consequences conveys that the action is appropriate. Model gender is less important in learning academic skills and strategies (Schunk, 1987).

Researchers have also explored the effects of exposing students to mastery and coping models (Schunk, 1987). Mastery models demonstrate faultless performance from the outset. Coping models initially have difficulty learning skills but through effort gradually improve their skills and eventually perform as well as mastery models. Research shows that children who have experienced previous learning difficulties may benefit more from observing coping models (Schunk & Hanson, 1985). Such children may perceive themselves as more similar in competence to coping models, which can raise their self-efficacy and motivation for learning.

Viewing one's own performances, or *self-modeling*, can facilitate learning and motivation. In a study by Carroll and Bandura (1982), adults viewed models performing a motor skill, then attempted to reproduce it. Performances of some learners were taped and learners were allowed to watch this concurrent visual feedback while performing. Visual feedback given before learners had formed a mental model of the skill had no effect on performance; however, once learners had formed such a mental model, the visual feedback enhanced their production of the skill. The self-modeled feedback presumably helped to reduce discrepancies between learners' mental models and actual performances.

Observational learning is enhanced when modeled displays contain explanations and demonstrations (*cognitive modeling*). Schunk (1981) compared the effects of cognitive modeling with those of didactic instruction on children's long-division self-efficacy and achievement. Children who lacked division skills received instruction and practice over

sessions. Cognitive-modeling children observed an adult model explain and demonstrate division solution strategies while applying them to problems. Didactic-instruction children received written instructional material that explained and demonstrated the operations. Compared with didactic instruction, cognitive modeling promoted division achievement and accuracy of perceived division capabilities (i.e., self-efficacy was better aligned with actual skills).

Symbolic Processes

In addition to their capability for vicarious learning, individuals possess the capacity for symbolic representations, which involve language, mathematical and scientific notation, iconography, and cognition, and which help people adapt to and alter their environments (Bandura, 1986). They use symbolic processes to interpret actions and outcomes in their lives and to guide their future actions. Because of the human capacity to symbolize, people do not simply react to events in their lives but rather generate new courses of actions for solving problems. Symbolic processes also foster communications with others (e.g., in person, on the phone, electronically, in writing), which lead to further learning.

Self-Regulatory Processes

Social cognitive theory assigns a prominent role to self-regulatory processes (Bandura, 1986; Zimmerman, 2000). *Self-regulation* refers to the processes that individuals use to personally activate and sustain behaviors, cognitions, and affects, which are systematically oriented toward the attainment of goals (Zimmerman, 2000). Prior to embarking on a task, people set goals and determine which strategies to use. They then regulate their behaviors to conform to their internal standards and goals. As they work on tasks, they assess their progress toward their goals and decide whether to continue or alter their strategies. During breaks and when tasks are complete, they reflect on their experiences, seeking to make sense of them and to determine what their next steps should be. As they reflect on what they have done, their beliefs that they have learned and made progress strengthen their self-efficacy and motivate them to continue learning. We elaborate on these motivational processes next.

Motivational Processes

The preceding sections show how observational, symbolic, and self-regulatory processes can have motivational effects on individuals. Among the most

Table 2.2. Key Social Cognitive Motivational Processes

• Goals and self-evaluations of progress
• Outcome expectations
• Values
• Social comparisons
• Self-efficacy

critical are goals and self-evaluations of progress, outcome expectations, values, social comparisons, and self-efficacy (Table 2.2). These processes are covered in the following sections.

Goals and Self-Evaluations of Progress

Goals, or what people are consciously trying to attain, involve important symbolic and self-regulatory processes that people use to instigate and sustain actions. Initially, people must make a commitment to attempt to attain goals because goals do not affect behavior without commitment (Locke & Latham, 2002). As persons work on a task, they compare their current performance with their goals. Positive self-evaluations of progress strengthen self-efficacy and sustain motivation. A perceived discrepancy between present performance and the goal may create dissatisfaction, which can raise effort. Goals motivate people to expend effort necessary and persist at the task (Locke & Latham, 2002). Greater effort and persistence typically lead to better performance. Goals also help to direct people’s attention to relevant task features, behaviors to be performed, and possible outcomes, and they can affect how people process information.

Although goals are important motivational processes, their effects depend on their properties: specificity, proximity, and difficulty. Goals that include specific performance standards are more likely to activate self-evaluations of progress and enhance motivation and learning than are general goals (e.g., “Do your best;” Bandura, 1986). Specific goals indicate the amount of effort needed to succeed, and evaluating progress toward specific goals is straightforward. Goals also are distinguished by how far they project into the future. Proximal, short-term goals enhance motivation and learning better than do distant, long-term goals, because it is easier to determine progress toward goals that are closer at hand (Bandura & Schunk, 1981).

Goal difficulty, which refers to the level of task proficiency required as assessed against a standard,

influences the amount of effort that people expend. In general, people work harder to attain goals perceived to be difficult than goals thought to be easier; however, perceived difficulty and motivation do not bear an unlimited positive relation to one another. Goals that people believe are overly difficult do not motivate because people hold low self-efficacy for attaining them. The opposite may also be true. Although people may feel efficacious for attaining goals perceived as very easy, these goals may not motivate because people often procrastinate in attempting them.

Another distinction can be made between learning and performance goals. A *learning goal* refers to what knowledge, behavior, skill, or strategy students are to acquire; a *performance goal* refers to what task students are to complete. These goals can have differential effects on achievement behaviors (Anderman & Wolters, 2006). Learning goals focus students’ attention on processes and strategies that help them acquire competence and improve their skills. Focusing on knowledge and skill acquisition motivates behavior and sustains attention to important features. Students in pursuit of a learning goal are apt to feel self-efficacious for attaining it and be motivated to expend effort, persist, and use effective learning strategies. Self-efficacy is substantiated as they work toward their goal and assess their progress (Schunk, 1996).

In contrast, performance goals focus attention on completing tasks. They may not highlight the importance of the processes and strategies underlying task completion or raise self-efficacy for learning. As students engage in a task, they may be less likely to determine their progress by comparing their present and past performances. Performance goals can lead to social comparisons with the work of others to determine progress. These comparisons can lower self-efficacy among students who experience learning difficulties, which adversely affects motivation and learning.

Research supports these hypothesized effects of learning and performance goals (Anderman & Wolters, 2006). For example, Schunk (1996) conducted two studies in which elementary children with low fraction skills received instruction and practice on fractions over sessions. Children worked under conditions involving either a goal of learning how to solve problems or a performance goal of merely solving them. In the first study, half of the students in each goal condition completed a self-evaluation at the end of each instructional session in which they evaluated their progress in

learning to solve the types of problems covered during that session. The learning goal with or without self-evaluation and the performance goal with self-evaluation led to higher motivation, self-efficacy, and achievement. In the second study, all students evaluated their learning progress at the end of the last instructional session. The learning goal led to higher motivation and achievement outcomes than did the performance goal.

Schunk and Ertmer (1999) conducted two studies with college undergraduates as they worked on computer projects over sessions. Students received a goal of learning computer applications or a goal of performing them. In the first study, half of the students in each goal condition evaluated their learning progress midway through the instructional program. The learning goal led to higher self-efficacy, self-judged progress, and self-regulatory competence and strategy use. The opportunity to self-evaluate progress promoted self-efficacy. In the second study, students in the self-evaluation condition assessed their progress after each instructional session. Frequent self-evaluation produced comparable results when linked with a learning or performance goal. These results suggest that infrequent self-evaluation of one's progress complements learning goals, but multiple self-evaluations can outweigh the benefits of learning goals and raise motivation and achievement outcomes.

Outcome Expectations

Outcome expectations are beliefs about the expected outcomes of actions. They can refer to external outcomes, such as "If I study hard, I should do well on the test." They also can refer to internal outcomes (e.g., "If I study hard, I will feel good about myself"), and to progress in learning (e.g., "If I study hard, I will learn more"). People form outcome expectations about the likely consequences of given actions based on personal experiences and observations of models (Bandura, 1986; Schunk & Zimmerman, 2006). Outcome expectations are a source of motivation. Individuals act in ways they believe they will be successful and attend to models whom they believe will teach them valued skills.

Outcome expectations can sustain behaviors over long periods when people believe their actions will eventually produce desired outcomes. Students who hold a sense of self-efficacy for succeeding and believe that their actions will result in positive outcomes are motivated to continue working even when progress occurs slowly. Conversely, those whose self-efficacy is weaker may, when they encounter difficulties,

work lackadaisically or give up readily. This situation can be demotivating; students may believe that positive outcomes will result but that they personally lack the self-efficacy to motivate themselves to continue. For example, they may believe that if they studied hard they would do well on the test, but they may doubt their self-efficacy to study hard.

Shell, Murphy, and Bruning (1989) obtained evidence of the influential role of outcome expectations. College students completed measures of self-efficacy and outcome expectations for reading and writing, as well as reading and writing achievement tests. For the self-efficacy assessment, students judged their competencies for performing various reading and writing tasks. For the outcome expectation measure, students judged the importance of reading and writing skills for achieving life goals, such as getting a job, being financially secure, and being happy.

Self-efficacy and outcome expectations related positively to achievement in both domains, although the relations were stronger for reading than for writing. In both domains, self-efficacy related more strongly to achievement than did outcome expectations, although the latter results were significant and added to the prediction of achievement. This study also found that self-efficacy and outcome expectations in each domain related significantly to achievement in the other domain, which suggests that improvements in students' self-efficacy and outcome expectations in one literacy area may generalize to other areas.

Values

Values are individuals' perceptions of the importance and utility of learning and acting in given ways. The role of values in motivation has been explored extensively by achievement motivation researchers (Eccles, 2005; Wigfield & Eccles, 2002; Wigfield, Tonks, & Eccles, 2004). Values enter prominently in a social cognitive account of motivation (Bandura, 1997). People who value attaining a sense of agency believe that they can exert a significant degree of control over important elements in their lives and are motivated to do so.

Individuals act in ways to bring about the outcomes they value and avoid actions leading to outcomes that are inconsistent with their values. They are motivated to learn when they deem that learning in a given area is important. Students who value mathematics may do so for various reasons, such as because they want to become mathematics teachers or because they believe that mathematics has many

uses in everyday life. Valuing mathematics may lead them to take more mathematics courses and expend greater effort to succeed.

Investigations by achievement motivation researchers have shown that values and expectancy beliefs such as self-efficacy relate positively to students' achievement. When both expectancy beliefs and values are used to predict achievement, expectancy beliefs are significant predictors, whereas values are not. In contrast, values are better predictors of students' intentions to take future courses and actual enrollment in those courses than are expectancy beliefs (Wigfield & Eccles, 2002). Thus, values seem most important as contributors to individuals' choices, which are key motivational outcomes.

Social Comparisons

Given its emphasis on learning from the social environment and reciprocal interactions among personal, behavioral, and social/environmental variables, social cognitive theory underscores the importance of *social comparisons*, which refer to the process of comparing ourselves with others (Wheeler & Suls, 2005). Although people often compare their performances with objective standards, they also socially evaluate their capabilities, especially when objective standards are unclear or unavailable. Comparisons indicating that one is improving or more competent than others can raise self-efficacy and motivation; comparisons that result in negative self-evaluations can diminish these outcomes.

The most accurate self-evaluations arise from comparisons with others whom people believe are similar to themselves in the particular ability or characteristic being evaluated. The more alike observers are to models, the greater the probability that similar actions by observers will produce comparable results (Schunk, 1987). Model-observer perceived similarity in competence can improve learning (Braaksma, Rijlaarsdam, & van den Bergh, 2002). Observing similar others succeed can raise observers' self-efficacy and motivate them to try the task. Similarity may be especially influential with persons who have experienced difficulties and possess self-doubts about performing well.

Although social comparisons can motivate individuals, their effects are not automatic. Among elementary school children, Schunk (1983a) found that providing children with social comparative information about how their performances compared with those of others promoted their motivation but that pursuing goals enhanced their self-efficacy. Giving children both goals and comparative information led

to the best learning. Schunk (1983b) showed that difficult goals raised children's academic motivation more than easier goals, that persuasive self-efficacy feedback (e.g., "You can work 25 problems") raised self-efficacy more than feedback indicating how children's performances compared to those of peers, and that difficult goals plus persuasive feedback led to the highest achievement.

The effects of social comparisons on self-efficacy and motivation depend on the abilities of the comparison peers. Guay, Boivin, and Hodges (1999) found that the relation between children's perceived competence (analogous to self-efficacy) and achievement was stronger when best friends' achievement was low than when it was high. Students' social comparisons with close friends' achievement may make students' own performances look worse than they really are. In contrast, children may assess their capabilities more accurately when they have low-achieving friends because they rely less on social comparison and more on objective assessments of their progress and performances.

Thus, it seems that social comparisons can enhance motivation but not necessarily self-efficacy or learning. Social comparisons that focus students on the accomplishments of similar and average peers imply that they, too, are average and therefore have no reason to feel highly self-efficacious. Self-efficacy may decline when students socially compare themselves to high-achieving peers. Self-efficacy and motivation may benefit more from providing students with objective information indicating that they are making learning progress without referring to peers' accomplishments.

Self-Efficacy

Self-efficacy is a critical variable affecting learning and motivation (Bandura, 1997). This section discusses how individuals develop and alter their self-efficacy, the consequences of self-efficacy, research on self-efficacy in achievement situations, and collective self-efficacy.

Sources of Self-Efficacy Information

People acquire information to assess their self-efficacy from four primary sources: their mastery experiences (interpretations of actual performances), vicarious (modeled) experiences, forms of social persuasion, and physiological indexes (Bandura, 1997; Table 2.3). One's actual performances constitute the most reliable information because they typically are interpreted as tangible indicators of one's capabilities (Schunk & Pajares, 2009; Usher & Pajares, 2008b).

Table 2.3. Informational Sources of Self-Efficacy

• Mastery experiences (interpretations of actual performances)
• Vicarious (modeled) experiences
• Forms of social persuasion
• Physiological indexes

Successful performances raise self-efficacy, whereas failures may lower it, although an occasional failure or success after many successes or failures should not have much impact.

The influence of actual performances on self-efficacy depends on numerous circumstantial factors such as task difficulty, effort expended, aid received, and preconceptions of one’s capabilities (Bandura, 1997). Consequently, the cognitive interpretations of the results of one’s actions, not the actions themselves, determine the influence of past performances on efficacy judgments. For example, meeting the minimum requirements for passing geometry may not boost the mathematics self-efficacy of a student who holds extraordinarily high personal standards and who longs to be an engineer. For another student, whose values and interests lie elsewhere, an average performance in geometry may boost self-efficacy and lead to continued motivation in mathematics classes. Failures can also serve an important function when they lead to better strategies that make self-efficacy more robust.

Individuals acquire much information about their capabilities through social comparisons with others (Bandura, 1997). Similarity to others is a cue for gauging one’s self-efficacy (Schunk, 1987). Observing similar others succeed can raise observers’ self-efficacy and motivation when they believe that if others can perform well, they can too. But a vicarious increase in self-efficacy can be negated by subsequent failure. Persons who observe similar peers fail may believe they lack the competence to succeed, which can negatively affect motivation.

In their daily school environments, students likely compare themselves to particular classmates who are engaged in similar learning activities. Surpassing one’s peers builds self-efficacy, whereas inferior performances lower it. Despite these tendencies, the influence of peer models on one’s self-efficacy cannot be reliably predetermined. A high-achieving, competitive student might get a self-efficacy boost from being outperformed by a classmate (Usher, 2009). On the other hand, a student who stands out

for superior performance among classmates might make external attributions (e.g., “I did well because the test was easy”) that leave self-efficacy relatively unchanged. Whether a vicarious experience raises or lowers self-efficacy depends on the models one selects for comparison, how similar the models are perceived to be, the models’ attitudes, and disparities between the observers’ and models’ achievement and progress.

Teaching practices can also increase the frequency with which students compare their performances to those of others. Schools create comparative structures when they group students according to academic ability levels as measured by achievement test scores or similar criteria. Such practices can send students a public message of their (in)efficacy. And because exposure to multiple skilled models sustains learning self-efficacy, students who find themselves among highly talented peers may reap long-term self-efficacy benefits, whereas those surrounded by less-skilled peers may harbor similar self-doubts. Students who internalize personal standards may be less prone to making unfavorable comparisons (Pajares, 2006).

Because of the human capacity for symbolism and forethought, people are capable of cognitively generating events that can serve as guides for action. Students are, therefore, partial creators of their modeled experiences. Through cognitive self-modeling, people are able to visualize themselves confronting and overcoming challenges (Bandura, 1997). Envisioning one’s academic success can raise self-efficacy, whereas imagining oneself failing lowers self-efficacy and can ensure the feared failure. The fact that this mode of cognitive influence has not been extensively examined need not suggest its impotence in changing self-efficacy. As William James remarked over a century ago, “The reaction due to things of thought is notoriously in many cases as strong as that due to sensible presences. It may be even stronger” (James, 1905, p. 53).

The third source of self-efficacy information on which individuals rely comes from the persuasive messages of others (e.g., “I know you can do this”; Bandura, 1997). But social persuasions must be credible for people to believe that success is attainable. Although positive feedback can raise individuals’ self-efficacy, the increase will not endure if they subsequently perform poorly (Schunk & Pajares, 2009). Factors that influence the persuasory punch of a message include source credibility, valence of the message, and frequency. A youngster’s self-efficacy is likely to suffer more from disparaging remarks than

from positive ones (Bandura, 1997). Students who hear frequent messages from multiple sources that they are incapable may come to believe that to be the case.

People are more likely to attend to social messages about their capabilities when they lack adequate knowledge of what is required to succeed in a particular domain. To be most effective and motivating, persuasive messages from others must be matched to the individuals' current skill level. Students are quick to dismiss lofty praise or empty inspirational mantras. Those who are most skilled at building students' self-efficacy couple positive feedback about students' capabilities with scaffolded tasks that build mastery (Evans, 1989).

Individuals also can acquire self-efficacy information from physiological and emotional reactions such as anxiety and stress (Bandura, 1997). Strong emotional reactions provide cues about anticipated success or failure. For example, a student who feels a crippling fear when heading to advanced chemistry may interpret that fear as a sign of personal inefficacy. When people experience negative thoughts and fears about their capabilities (e.g., feeling nervous when thinking about taking a test), those affective reactions can lower self-efficacy. On the other hand, positive affect or excitement in learning can motivate. A student who feels energized by challenging academic work likely enjoys a sense of self-efficacy for succeeding. When people notice their stress abating (e.g., feeling less anxious while taking a test), they may experience higher self-efficacy for performing well.

As we noted earlier, informational sources related to one's abilities do not affect self-efficacy automatically (Bandura, 1997). Individuals interpret the results of events, and these interpretations provide the impetus for upward or downward shifts in one's self-efficacy (Schunk & Pajares, 2009). People weigh and combine information from the various sources to form self-efficacy judgments. Many factors influence the ways in which students interpret and integrate this information when forming their self-efficacy and motivation-related beliefs. For some individuals, the accumulation of informational sources enhances self-efficacy. Other people tend to rely on information from one source more than from others. For example, in a study of the sources of academic self-efficacy among middle school students, girls and African American students seemed more attuned to social persuasions when forming their self-efficacy than did boys and White students (Usher & Pajares, 2006).

The influence of these sources of self-efficacy might also be multiplicative, in that two sources combine interactively. Students who have had few mastery experiences in a given domain may be more likely to rely on what others tell them than would students who have had ample opportunities for mastery (Usher, 2009). Beliefs in one's personal efficacy for learning might also follow a transformational experience. A meaningful individual encounter with a caring teacher might have a more profound influence on one's self-efficacy than a year's worth of school. A disparaging remark can also leave a lasting bruise on one's sense of efficacy and undermine subsequent motivation. We now turn to the many outcomes that are influenced by these important self-beliefs.

Effects of Self-Efficacy

Within a social cognitive system of triadic reciprocity, self-efficacy is hypothesized to influence behaviors and environments and in turn be affected by them (Bandura, 1986, 1997). Self-efficacy exerts its influence through cognitive, motivational, affective, and selection processes. Students who feel efficacious about learning should engage in thoughts and actions that improve their learning, such as setting goals, using effective learning strategies, monitoring their comprehension, evaluating their goal progress, and creating effective environments for learning. In turn, self-efficacy is influenced by the outcomes of one's behaviors (e.g., goal progress, achievement) and by input from one's environment (e.g., feedback from teachers, social comparisons with peers; Schunk & Pajares, 2009).

Despite its benefits, self-efficacy is not the only influence on behavior. No amount of self-efficacy will produce a competent performance when individuals lack the needed skills to succeed (Schunk & Pajares, 2009). Discussed earlier was the importance of other motivating factors such as outcome expectations and values (Bandura, 1997; Wigfield et al., 2004). Even learners who feel highly efficacious about their mathematical skills will not become mathematics majors in college if they do not value a career as a mathematician, and they typically engage in activities that they believe will result in positive outcomes and avoid actions that they believe may lead to negative outcomes. Nonetheless, given requisite skills, positive values, and outcome expectations, self-efficacy is a key determinant of individuals' motivation, learning, self-regulation, and achievement (Schunk & Pajares, 2009).

Self-efficacy can have diverse effects in achievement contexts (Bandura, 1997; Pajares, 1996; Schunk & Pajares, 2009; Table 2.4). It can influence various motivational outcomes, including choice of tasks, effort, and persistence. Individuals are apt to select tasks and activities at which they feel competent and avoid those at which they do not. Self-efficacy can affect how much effort people expend on an activity, how long they persist when they encounter difficulties, and their levels of learning and achievement. People with high self-efficacy tend to set challenging goals, work diligently, persist in the face of failure, and recover their sense of efficacy after setbacks. As a consequence, they develop competence. On the other hand, those with low self-efficacy may set easier goals, expend little effort to succeed, give up readily when they experience difficulties, and feel dejected after they encounter failure, all of which negatively affect skill acquisition.

Self-efficacy also influences one's level of self-regulation (Schunk & Pajares, 2009; Zimmerman & Cleary, 2009). Those with higher self-efficacy for learning set challenging goals, employ what they believe are effective strategies, self-monitor their learning goal progress, make strategy adjustments and seek help as needed, and create an effective work environment. As formal and informal learning environments become increasingly technological, one's capabilities to minimize distractions and find reliable information are at a premium. In turn, these activities result in better performance and higher self-efficacy for continued improvement. We next highlight some specific research findings on the effects and sources of self-efficacy.

Research Evidence

Researchers have explored the operation of self-efficacy in various domains (e.g., education, health, business) and among individuals differing in age, developmental level, and cultural background. This research has shown that self-efficacy is a strong predictor of individuals' motivation, achievement,

self-regulation, and life decisions in diverse contexts (Bandura, 1997; Klassen & Usher, 2010; Multon, Brown, & Lent, 1991; Pajares, 1997; Schunk & Pajares, 2009; Stajkovic & Luthans, 1998).

For example, much research shows that self-efficacy correlates with motivation, learning, and achievement (Schunk & Pajares, 2009). Using meta-analysis, Multon et al. (1991) found that self-efficacy was related to academic performance and accounted for 14% of the variance. Stajkovic and Luthans (1998) found that self-efficacy resulted in a 28% gain in work performance. Using path analysis, Schunk (1981) found that self-efficacy exerted a direct effect on children's mathematics achievement and persistence. Pajares and Kranzler (1995) found that mathematics self-efficacy had a direct effect on mathematics performance and mediated the influence of mental ability. Self-efficacy for self-regulated learning also predicts academic motivation, achievement, and continuation in school (Caprara et al., 2008; Usher & Pajares, 2008a).

Self-efficacy relates not only to task choice but also to career choice (Betz & Hackett, 1983). Social cognitive career theorists have demonstrated that basic social cognitive variables, including self-efficacy, outcome expectations, and goals, help explain career decision making and development (Brown & Lent, 2006). As learners grow and are given more choices over their decisions and activities, they are apt to select activities that involve capabilities they believe they can develop and turn away from areas in which they have doubts. The influence of self-efficacy on career development is partly mediated by perceived effort and persistence (Schunk & Pajares, 2009).

The relation of self-efficacy to effort and persistence is not always linear. In novel learning situations, students initially do not possess skills and must expend effort and persist to succeed. As skills develop, however, students should be able to attain the same level of performance with less effort in a shorter time. When this does not happen, self-efficacy may decline. Thus, if an advanced student believes she is capable in science but suddenly must exert a herculean effort to pass physics, she might begin to rethink her pursuit of a science-related career.

In addition to documenting the effects of self-efficacy, researchers have examined the influence of the four hypothesized sources on self-efficacy development. Mastery experiences have been shown to be the most powerful and consistent predictor of self-efficacy across academic domains and age levels

Table 2.4. Effects of Self-Efficacy

• Motivational outcomes (task choices, effort, persistence)
• Learning
• Achievement
• Self-regulation

(Usher & Pajares, 2008b). Scaffolding instruction to provide for frequent successes offers learners many opportunities to build a sense of self-efficacy in their capabilities. The relative predictive power of the other three sources has been variable across studies. For example, in their study of the sources of academic and self-regulatory efficacy beliefs of sixth-grade students, Usher and Pajares (2006) found that girls and African American students relied on the social persuasions of others when forming their confidence, whereas this source was not a significant predictor of boys' or White students' self-efficacy. Klassen (2004) also found that Indo-Canadian students reported greater reliance on vicarious experiences and social persuasions than did Anglo-Canadian students. Investigating the importance of social messages, whether transmitted through vicarious enactment or verbal persuasion, for various groups of learners remains an important area of inquiry for understanding how efficacy beliefs take root.

Experimental research has shown that instructional and social processes that convey information to students that they are making learning progress and becoming more competent raise self-efficacy, motivation, and achievement (Schunk & Pajares, 2009). Other instructional strategies for building students' self-efficacy include having students pursue proximal and specific goals, using social models in instruction, providing social comparative information indicating competence, and having students self-monitor and evaluate their learning progress (Schunk & Ertmer, 2000). A noncompetitive classroom climate can lower students' anxiety, which can lead to a more favorable evaluation of their own capabilities (Bandura, 1997).

Collective Efficacy

Researchers have explored the operation of *collective efficacy* beliefs, or individuals' beliefs about their collective capabilities to learn or produce desired actions (Bandura, 1997). Collective efficacy perceptions are not simply the sum or average of the self-efficacy of individual group members; rather, they reflect individuals' perceptions of the capabilities of the group as a whole. In educational settings, collective teacher efficacy denotes the perceptions of the faculty as a whole to influence student outcomes (Goddard, Hoy, & Woolfolk Hoy, 2000; Henson, 2002).

The role of collective efficacy beliefs on group motivation may depend on the level of organizational coupling (Henson, 2002). In units that are

loosely knit, collective efficacy beliefs may not predict outcomes well; rather, individual self-efficacy may be a better predictor. Conversely, in more tightly knit units—such as many elementary schools—the collective efficacy beliefs of teachers may be a better predictor of the efforts of the faculty as a whole to affect student learning (Henson, 2002).

The same four sources are important for the development of collective self-efficacy: performance attainments, vicarious experiences, social persuasion, and physiological indicators. Group members rely on what they know about the capabilities of each group member, as well as the group's collective capacity, when evaluating what they can do together (Bandura, 1997). When members work together successfully to implement changes, learn from one another and from other successful groups, receive encouragement for change from supervisors and others, and work together to cope with difficulties and alleviate stress, their beliefs about what they can do will be raised (Goddard, Hoy, & Woolfolk Hoy, 2004). Individuals who believe in their group's collective capabilities will be more motivated to work on the group's behalf, implement innovative ideas, and enact systemic change.

Educational research shows that collective self-efficacy is important for teachers' job satisfaction and motivation to remain in teaching. Researchers have found that teachers' collective self-efficacy bears a significant and positive relation to their job satisfaction in various contexts (Caprara, Barbaranelli, Borgogni, & Steca, 2003; Klassen, Usher, & Bong, 2010). Relatively less research attention has been given to the collective efficacy beliefs of students. Klassen and Krawchuk (2009) showed that the collective efficacy beliefs of early adolescents working in small groups became more strongly related to the group's success on an interdependent task over time. Perceived collective efficacy also depends on group members believing that others are working on their behalf. Consistent with Bandura's (1997) contention, however, even high self-efficacy will not lead to performance changes unless the environment in which groups function provides appropriate avenues for success.

Future Directions

Social cognitive theory offers a viable account of motivation, and researchers continue to test its predictions. But there remain several questions that should be addressed by investigators. In particular, research is recommended on the benefits of modeled observations, developmental appropriateness, and cross-cultural relevance.

Benefits of Modeled Observations

Individuals learn new skills and strategies by observing models. Modeled observations also motivate observers to improve their skills. In educational settings, use of models is apt to save instructional time as teachers or others can explain and demonstrate skills and strategies to be acquired. This also prevents students from learning inaccurately, as might happen if there were less structure.

Although models are important, their effects on students' self-efficacy and motivation are weaker than are those resulting from actual performance accomplishments. Researchers might explore how best to combine modeled demonstrations with learner practice to optimize motivational effects. In some situations, relatively little practice may be needed, but more is likely when skills to be learned are complex. Such research would contribute to clarifying how learners weigh and combine sources of self-efficacy information to arrive at self-efficacy judgments. For example, how is self-efficacy affected if models perform successfully but students then have difficulty when they practice? This type of research also would have instructional implications because it would suggest ways to effectively use instructional time to promote self-efficacy and motivation.

Technological innovations might facilitate this line of research. Computers and handheld devices make it possible for modeled experiences to be at students' fingertips. For example, researchers could use video recordings of models at varying skills levels to examine their influence on a diverse group of learners. An experimental design would enable varying of model characteristics such as similarity, proficiency, and degree of shared coping. Video playback of one's own performances could enhance students' ability for cognitive self-modeling as well. The changing nature and availability of technology make possible new and diverse modeling opportunities. If such videos of modeled skills prove to be effective, teachers could benefit from developing their own library of vicarious experiences for their students.

Developmental Appropriateness

Social cognitive theory emphasizes complex interactions among personal, social/environmental, and behavioral factors. This complexity leads to questions about the applicability of the theory to learners of all ages and developmental levels. For example, when assessing self-efficacy, individuals must weigh and combine information from the environment, their prior experiences, and their

perceptions of the present situation. Such complex cognitive processing may be beyond the capabilities of young children, which can diminish the predictive utility of self-efficacy. Furthermore, mentally processing information conveyed by models can be complex, as when models demonstrate problem-solving strategies.

Children can learn from observing models and make reasonably valid self-efficacy judgments (Bandura, 1986). Models for children are effective when their explanations and demonstrations are brief and restricted to specific skills. Self-efficacy assessments typically contain a restricted range of choices, and children are given practice to ensure that they understand the nature of the judgment process. Thus, although social cognitive principles are assumed to apply to learners at different developmental levels, researchers might explore what constraints developmental factors place on applying these principles.

Longitudinal designs that track changes in learners' self-efficacy and motivation over time would help researchers understand the influence of developmental stages on efficacy appraisals. Such designs could also target changes in the efficacy-related information that students perceive. Multi-level modeling techniques could help document how self-efficacy levels fluctuate among groups of students (e.g., between grades 3 and 5) and could examine predictors of individual students' self-efficacy growth trajectories. Despite the costs of conducting longitudinal research, such designs will be able to clarify important questions related to developmental shifts in the sources and effects of self-efficacy that have not been clearly answered by cross-sectional designs.

Cross-Cultural Relevance

Pajares (2007) called for a careful consideration of cultural context in the investigation of academic motivation in general and self-efficacy in particular. Because the relation between self-efficacy, other motivation variables, and achievement varies in important ways across cultural groups, researchers should use caution when generalizing research results to other contexts. As Pajares (2007) noted, "Research findings must be carefully understood as being bounded by a host of situated factors" (p. 30), which limit what is known about a given variable.

Most social cognitive research relevant to motivation has been conducted with individuals in Western societies (Klassen & Usher, 2010). Fortunately this situation is changing as researchers are testing social cognitive ideas in settings globally.

As a research topic, self-efficacy has much international appeal, with the resulting increase in research in different cultures. While cross-cultural research has yielded differences (McInerney, 2008), overall the principles espoused by social cognitive theory have shown themselves to be cross-culturally relevant. Additional investigations will determine whether the motivational processes postulated by social cognitive theory operate consistently in diverse societies.

Conclusion

Social cognitive theory stresses learning from the social environment. The conceptual focus of Bandura's (1986) social cognitive theory postulates reciprocal interactions among personal, behavioral, and social/environmental factors. Social cognitive researchers have investigated the operation of vicarious, symbolic, and self-regulatory processes, in the various ways that individuals interact with their environments and one another.

A key point underlying social cognitive theory is that persons are motivated to develop a sense of agency for being able to exert a large degree of control over important events in their lives. Among the influential variables affecting motivation are goals and self-evaluations of progress, outcome expectations, values, social comparisons, and self-efficacy. Important questions remain to be addressed by researchers, which will further refine social cognitive theory and expand its applicability to motivation.

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